

**COMPUTER VISION – CO3 ASSESSMENT TOOL 2 CASE
STUDY**

Course Name: Computer Vision

Course Code: ITA05

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CASE STUDY 1 – Preprocessing and Feature Detection

Problem Statement

The vision system captures images under varying lighting conditions, causing inconsistent feature detection. Uneven illumination and noise can create false keypoints and hide weak features.

Proposed Solution

Apply contrast enhancement and noise reduction before feature detection. CLAHE improves local contrast and Gaussian filtering suppresses high-frequency noise; SIFT can then detect more stable features.

Methodology / Algorithm

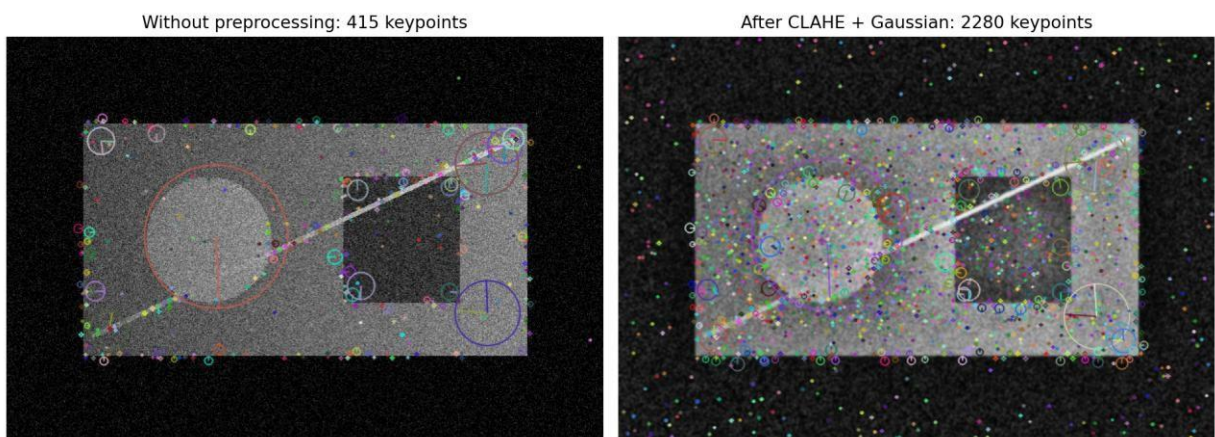
1. Convert the image to grayscale.
2. Apply CLAHE for local contrast enhancement.
3. Apply Gaussian filtering to reduce noise.
4. Run SIFT feature detection.
5. Compare keypoint distribution before and after preprocessing.

Dataset Used

A synthetic image with geometric structures was modified using uneven illumination and Gaussian noise. The supplied question paper does not specify a named dataset.

Results with Screenshot

The screenshot compares SIFT keypoints before and after preprocessing. The processed image gives a cleaner and more stable feature distribution.



Conclusion

Preprocessing improves feature stability by reducing noise and compensating for illumination differences. Therefore, preprocessing should precede feature detection when lighting varies.

CASE STUDY 2 – Edge vs Corner Detection

Problem Statement

Edge detection identifies boundaries but may not provide distinctive localized points for matching. In an image containing straight edges and textured regions, this can cause matching or tracking to fail.

Proposed Solution

Use corner detection such as Shi-Tomasi or Harris because corners have strong intensity changes in two directions and provide localized, distinctive keypoints.

Methodology / Algorithm

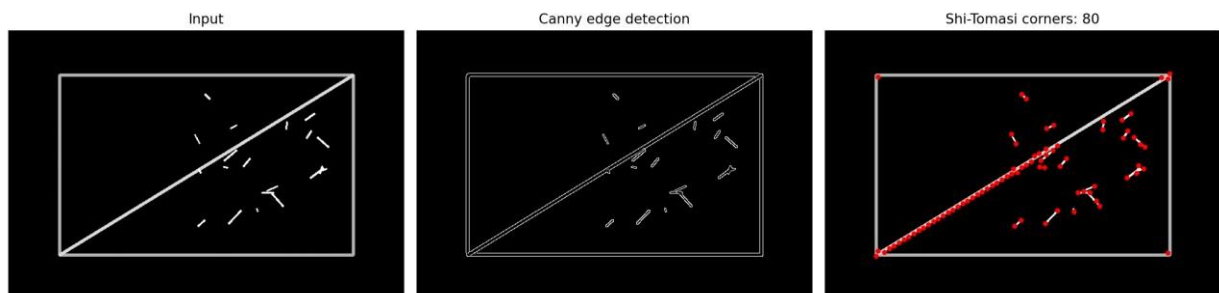
6. Convert the image to grayscale.
7. Apply Canny edge detection.
8. Observe the long continuous boundaries produced by edge detection.
9. Apply Shi-Tomasi corner detection.
10. Use the localized corners for matching or tracking.

Dataset Used

A synthetic image containing straight edges and a textured region was used for demonstration. The supplied question paper does not prescribe a particular dataset.

Results with Screenshot

The screenshot shows continuous boundaries from Canny and discrete corner points from Shi-Tomasi. The latter are better localized for correspondence.



Conclusion

Edges are useful for boundary extraction, but corners are more suitable when distinctive point correspondences are required for matching and tracking.

CASE STUDY 4 – Feature Matching under Transformations

Problem Statement

The same object is captured at different scales and orientations. Traditional matching is sensitive to these changes, so corresponding features may not be recognized.

Proposed Solution

Use a scale- and rotation-invariant method such as SIFT. Match its descriptors using nearest-neighbour matching and reject ambiguous pairs with the ratio test.

Methodology / Algorithm

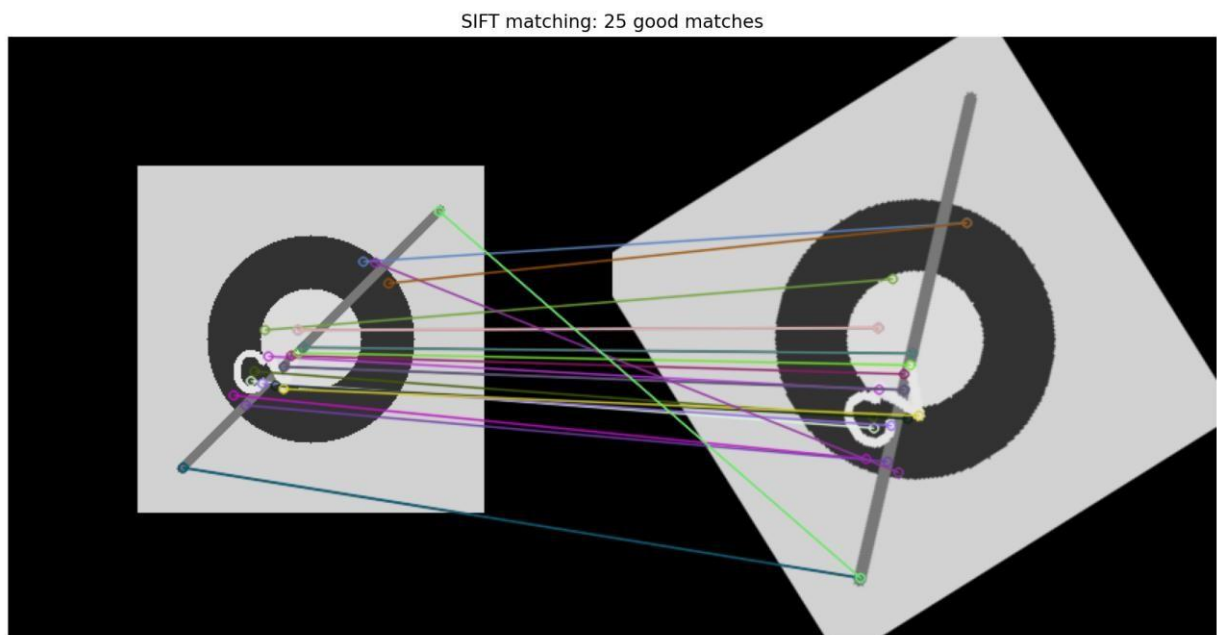
11. Detect SIFT keypoints and descriptors in both images.
12. Represent local structures using SIFT descriptors.
13. Match descriptors using a brute-force nearest-neighbour matcher.
14. Apply Lowe's ratio test.
15. Visualize the retained correspondences.

Dataset Used

A synthetic object image and a transformed version rotated by about 32° and scaled by $1.35\times$ were used. The supplied question paper does not specify a named dataset.

Results with Screenshot

The screenshot shows successful SIFT correspondences despite scale and orientation changes.



Conclusion

Scale- and rotation-invariant descriptors are robust to geometric transformations, making SIFT suitable for matching the same object under different scales and orientations.